

Bangladesh National Building Code (BNBC)

Framework, Application & Real-Life Impacts
and Ethics

Lecture Objectives

- Understand BNBC purpose and structure
- Learn key safety and design provisions
- Apply BNBC in real-life construction
- Explore compliance and case examples

What is BNBC?

BNBC (Bangladesh National Building Code) provides **minimum standards** for:

- Building design
- Construction
- Safety
- Maintenance

Published first in 1993 by Housing and Building Research Institute (HBRI) under Ministry of Housing and Public Works (MoHPW).

Revised in 2020 to incorporate updated seismic, fire, and environmental standards.

Objectives of BNBC

- Ensure structural safety and stability
- Protect life and property from hazards (fire, earthquake, wind, etc.)
- Promote energy-efficient, sustainable building practices
- Maintain uniformity in design and construction practices across the country

BNBC Framework

BNBC 2020 is divided into **10 parts**, each addressing specific aspects:

Part	Title	Focus Area
Part 1	Administration	Responsibilities, permits, inspections
Part 2	Site Preparation & Environment	Site safety, sanitation, soil study
Part 3	General Building Requirements	Building occupancy, height, setbacks
Part 4	Structural Design	Load calculation, material design
Part 5	Fire Protection	Fire resistance, exits, hydrants
Part 6	Building Materials	Standardization, quality testing
Part 7	Construction Practices	Workmanship, safety, inspection
Part 8	Plumbing & Sanitation	Water supply, drainage
Part 9	Electrical & Mechanical	Wiring, HVAC, safety systems
Part 10	Sustainability & Energy	Green design, energy codes

Part 1: Administration

Scope and Purpose

- Outlines roles, responsibilities, and procedures for:
 - Architects
 - Engineers
 - Builders
 - Government authorities
- Includes permit system, inspection rules, and enforcement mechanisms.

Real-Life Example

When a developer in Dhaka wants to construct a 10-storey building:

- Must submit drawings to **RAJUK** for approval.
- Structural engineer must sign compliance with BNBC.
- Site inspections are done at each construction phase

Part 2: Site Preparation and Environmental Considerations

Scope

Ensures building site is safe, accessible, and environmentally compatible.

Requires:

- Soil bearing capacity tests
- Flood level clearance
- Proper drainage
- Tree and ecosystem preservation

Real-Life Example

Metro Rail depots and Padma Bridge service areas followed BNBC Part 2 to ensure:

- Flood resilience
- Drainage and slope stability
- Environmental protection

Part 2: Site Preparation and Environmental Considerations



Ethical Practice



Conduct soil investigation before design.



Avoid building in areas that are easily flooded or on land that can slide or collapse



Manage construction waste properly.



Unethical Practice



Dumping construction waste into rivers or drains.



Ignoring problems like polluted underground water or soil being washed away from slopes.



Constructing without environmental clearance.



Part 3: General Building Requirements

◆ Scope

- Code explains how buildings are categorized and controlled, including:
 - ✓ **Occupancy classification** – the type of use (like residential, commercial, or industrial).
 - ✓ **Setback rules** – how far a building must be from the property boundary or road.
 - ✓ **Floor Area Ratio (FAR)** – the ratio of total floor area to the size of the land.
 - ✓ **Height limits** – the maximum height a building is allowed to have.
- Also ensures ventilation, natural lighting, and accessibility (for persons with disabilities).



Real-Life Example

- Residential apartments in Mirpur follow FAR as per BNBC for density control.
- Educational buildings like universities must have open space and accessible routes.



Part 3: General Building Requirements



Ethical Practice

- ✓ Keep the proper distance (setbacks) from property lines and follow the allowed building size (FAR).
- ✓ Ensure accessibility so that everyone, including people with disabilities, can enter and move around easily — by using ramps, lifts, etc.
- ✓ Make stairs and corridors wide enough for safe and comfortable movement of people.



Unethical Practice

- ✗ Building extensions into roads or setbacks.
- ✗ Reducing corridor widths to get more rooms or space to rent, which is not safe.
- ✗ Ignoring accessibility features for disabled persons.



Part 4: Structural Design

◆ Scope

Specifies standards for:

- Different types of loads a building must handle — such as its own weight (dead load), people and furniture (live load), wind, and earthquake (seismic) forces.
- Design and strength requirements for materials like concrete, steel, and masonry (brick/block).
- Proper design of foundations and retaining walls to ensure buildings are stable and safe



Real-Life Example

- Padma Bridge approach buildings used deep pile foundations according to BNBC load-bearing requirements.



Part 4: Structural Design

Ethical Practice

- ✓ Perform accurate calculations to make sure the structure is strong and safe.
- ✓ Use only good-quality materials that are tested and approved
- ✓ Check the structure by load testing when needed to confirm it can safely carry the expected weight.

Unethical Practice

- ✗ Using substandard steel or cement to cut costs.
- ✗ Ignoring load combinations i.e. different forces (like weight, wind, and earthquake) act together, which can make a building unsafe
- ✗ Copying design from other buildings without site adaptation.

Part 5: Fire Protection

◆ Scope

- Defines sets rules for fire safety in buildings, including How long walls, floors, and staircases should resist fire, Minimum width for exits so people can escape safely and installation of fire hydrants and alarm systems to detect and fight fires.

Real-Life Example

FR Tower fire (2019): one staircase and no sprinklers → tragic result.

- BNBC requires:
 - Minimum 2 fire exits
 - Fire alarms
 - Smoke detection and fireproof doors

Part 5: Fire Protection

Ethical Practice

-  Follow BNBC fire standards strictly.
-  Install alarms and hydrants in all floors.
-  Regularly inspect and maintain fire safety equipment.

Unethical Practice

-  Blocking fire exits with furniture or storage.
-  Using flammable materials in escape routes.
-  Tampering with fire alarms to save electricity.



Part 6: Building Materials

◆ Scope

- Ensures all materials meet BDS (Bangladesh Standards) or international equivalents (ASTM, ISO).
- Covers:
 - Cement, steel, bricks, timber
 - Testing procedures and certification



Real-Life Example

Government construction projects (e.g., Rooppur NPP township) require certified material testing reports from BSTI.



Part 6: Building Materials



Ethical Practice



Use only certified materials from approved suppliers.



Store materials carefully so they don't get damaged or spoiled



Regularly test material samples to make sure they still meet quality standards.



Unethical Practice



Mixing cement with sand to save cost which weakens the structure.



Using steel of unknown quality or strength, which is unsafe.



Making fake test certificates to make materials appear approved, which is illegal and dangerous.



Part 7: Construction Practices

◆ Scope

- Covers
 - ✓ On-site supervision to ensure work is done correctly and safely
 - ✓ Safety practices to protect workers and the public
 - ✓ Workmanship and quality checks to make sure construction meets standards.
- Contractor's and engineer's roles clearly defined.



Real-Life Example

Metro Rail and Karnaphuli Tunnel followed strict site supervision and quality control checks per BNBC.



Part 7: Construction Practices



Ethical Practice



Provide safety equipment (helmets, gloves, harnesses) for all workers.



Record daily work and inspections to track progress and issues.



Follow proper procedures for concrete curing (harden properly) and using formwork (shuttering) to ensure strong construction.



Unethical Practice



Hiring unskilled workers without supervision.



Ignoring safety helmets or harnesses.



Rushing concrete curing for faster progress.

Part 8: Plumbing and Sanitation

◆ Scope

- Provides standards for:
 - Water supply & distribution
 - Drainage and sewage disposal
 - Septic tank design
 - Rainwater harvesting

Real-Life Example

Modern hospitals and universities in Dhaka use dual plumbing systems for potable and recycled water as per BNBC.



Part 8: Plumbing and Sanitation



Ethical Practice



Use sanitary-grade materials i.e., choose pipes, fittings, and other materials that won't contaminate water and are safe for health.



Make sure pipes and drains have the right slope so water flows properly.



Keep sewage and drinking water separate to avoid contamination.



Unethical Practice



Direct discharge of waste into open drains.



Using low-quality pipes prone to leakage.



Ignoring maintenance of septic systems i.e. not taking care of septic tanks and sewage systems, which can lead to blockages, leaks, or contamination of water and soil.

Part 9: Electrical and Mechanical Systems

Scope

- Focuses codes on Safe electrical wiring and equipment, Heating, ventilation, and air conditioning (HVAC) systems, Lift (elevator) installations, and Lightning protection to prevent damage.
- Aligns with IEC and BDS codes.

Real-Life Example

Commercial buildings in Gulshan require electrical inspection certificates before occupation.

Part 9: Electrical and Mechanical Systems

Ethical Practice

-  Install proper safety devices like circuit breakers to prevent electrical accidents.
-  Keep emergency system wiring separate from regular circuits for safety.
-  Have licensed electricians regularly inspect electrical systems to ensure they are safe and working correctly.

Unethical Practice

-  Overloading circuits or skipping earthing.
-  Using counterfeit electrical fittings i.e., using fake or low-quality electrical parts that pretend to be real, such as switches, sockets, circuit breakers, or wires that are not certified.
-  Ignoring maintenance schedules.



Part 10: Sustainability and Energy Efficiency

◆ Scope

Introduced in BNBC 2020 to promote green and low-carbon construction.

- Covers:
 - Energy-efficient lighting
 - Renewable integration (solar rooftop)
 - Rainwater harvesting
 - Waste management



Real-Life Example

BRAC University's new campus in Dhaka designed under green building principles:

- Solar PV
- Daylight optimization
- Water recycling



Part 10: Sustainability and Energy Efficiency



Ethical Practice



Adopt renewable technologies when possible.



Encourage waste recycling during construction i.e., promote reusing or recycling construction waste (like concrete, wood, metal, and packaging) instead of throwing it away, to reduce environmental impact.



Comply with national environmental policy.



Unethical Practice



Wasting construction materials.



Ignoring energy audits or green certification.



Making fake sustainability data to gain approval.